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Batch: Data Analytics Dec live Batch

Assignment: Hypothesis Testing

## Assignment

**Question 1: What is a null hypothesis ( $H_0$ ) and why is it important in hypothesis testing?**

**Answer:**

The null hypothesis ( $H_0$ ) is a statement that assumes no effect, no difference, or **no change**. Null hypothesis ( $H_0$ ) represents the current or **default assumption**.

**Importance:**

- It provides a starting point for testing
- Helps decide whether observed results are due to chance or not

**Example:**

$H_0$ : Average complaint resolution time is **10 minutes**.

**Question 2: What does the significance level ( $\alpha$ ) represent in hypothesis testing?**

**Answer:**

The significance level ( $\alpha$ ) is the **probability of rejecting a true null hypothesis**.

Common values:

- **$\alpha = 0.05$**  (5%)

**Question 3: Differentiate between Type I and Type II errors.**

**Answer:**

**Type I** error is **rejecting a true null hypothesis**

while **Type II** error is **failing to reject a false null hypothesis**.

**Question 4: Explain the difference between a one-tailed and two-tailed test. Give an example of each.**

**Answer:**

**One-tailed Test**

- Tests only one direction
- Example: "Has productivity increased?"

$H_1$ : Mean  $> \mu$

**Two-tailed Test**

- Tests both directions
- Example: "Has the mean changed?"

$H_1$ : Mean  $\neq \mu$

**Question 5: A company claims that the average time to resolve a customer complaint is 10 minutes. A random sample of 9 complaints gives an average time of 12 minutes and a standard deviation of 3 minutes. At  $\alpha = 0.05$ , test the claim.**

**Answer:**

**Given:** Sample Mean( $\bar{x}$ ) = 12 min

Critical Mean( $\mu_e$ ) = 10 min

Sigma = 3 min

significance level ( $\alpha$ ) = 0.05

n = 9

**Step 1: null hypothesis** : Customer complain solved in 10 min.

**Alternative hypothesis**: Customer complain not solved in 10 min.

**Step 2: Sample size ( $n$ ) < 30 that means t test**

### Step 3: Apply Formula of t test

$$(\bar{x} - \mu) / (\text{sample std} / \sqrt{n})$$

$$(12-10)/3/\sqrt{9}) = 2$$

### Step 4: Critical Value

Type is two tail.

$$df = n - 1 = 8$$

$$t_{0.05} \approx 2.306$$

### Step 4: Decision

Here , Calculated  $t < \text{Critical } t$

$$|t| = 2 < 2.306$$

since calculated  $t < \text{critical } t$ , accept null hypothesis reject alternate hypothesis

**Question 6: When should you use a Z-test instead of a t-test?**

**Answer:**

**Z test . is used when**

1. Sample size  $(n) \geq 30$
2. Population std should be known
3. data is roughly normally distributed

**T test is ued when**

1. Sample size  $(n) < 30$ .
2. Population std is unknown

